Predicting Free vs Paid Games on Steam

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Summary

This project uses machine learning to predict whether a Steam game is free-to-play or paid based on its attributes. Using a snapshot of approximately 5,000 Steam games collected in August 2025 (vintagedon (2025)), we train a cross-validated logistic regression classifier. The final model achieves 95.9% accuracy and a ROC AUC of 0.923 on the held-out test set, substantially outperforming a majority-class baseline of 79.4%.

Introduction

Steam is the world's largest PC game distribution platform, with over 10,000 annual releases (Statista Research Department (2026)) and around 20 million daily active users (Steam (n.d.)). Being able to predict whether a game will be free or not helps developers make pre-launch pricing decisions, helps publishers evaluate projects, and helps researchers study quality signals in large content marketplaces.

The dataset used is the Steam 2025 5K Games Dataset (vintagedon (2025)): a gzip-compressed JSON snapshot of approximately 5,000 Steam game records taken on August 31, 2025. Key fields include `steam_appid`, `name`, `is_free` (the target variable), `genres`, `categories`, `platforms`, `release_date`, `required_age`, and `developer`.

Specifically, we ask:

1. What game attributes differ most between free-to-play and paid titles?
2. Can a machine learning classifier accurately distinguish free and paid games using only pre-launch observable features?
3. Which features contribute most to that classification?

Methods

The R programming language (R Core Team (2024)) and the following R packages were used to perform the analysis: tidyverse (Wickham et al. (2019)), caret (Kuhn (2024)), pROC (Robin et al. (2011)), janitor (Firke (2023)), lubridate (Grolemund and Wickham (2011)), scales (Wickham, Pedersen, and Seidel (2023)), patchwork (Pedersen (2024)), jsonlite (Ooms (2023)), and Quarto (Allaire et al. (2022)).

Overview of Methodology

Our questions about the differences between free and paid games indicate that this is a **classification** problem. Thus, **logistic regression** will be our model of choice.

First, data was cleaned to simplify column names, recode variables to the appropriate type, and some exploratory data analysis was performed, leading to the discovery that the dataset was imbalanced. The imbalance is not too severe, but we must be aware of it when splitting our data and interpreting our results. With that in mind, we split the data into stratified train and test sets.

The logistic regression model was trained with 5-fold CV to optimise ROC AUC.

Data Wrangling & Cleaning

Data was loaded, cleaned, and feature-engineered using `scripts/02_data-preprocessing.R`. Column names were simplified, `is_free` was recoded as a two-level factor, platform flags were cast to logical and summed into `platform_count`, and the top 15 most-frequent game categories were one-hot encoded as indicator columns. Missing values were imputed with sensible defaults, keeping data loss below 10%.

Data Validation

Data validation was carried out by a pointblank pipeline. The pipeline checks that the following are as expected: - source file format - required columns - empty observations - missingness - data types - duplicates - value ranges - category levels - target distributions

Optional developer/publisher metadata missingness is recorded as a warning because those fields are not model predictors.

Correlation diagnostics are run only on the training split after stratified sampling. This avoids using held-out test-set relationships to guide modelling decisions. After running correlation diagnostics, we decided to remove two engineered predictors (cat_game_demo and cat_downloadable_content)that were perfectly or almost perfectly(99.95%) collinear.

Exploratory Data Analysis

The dataset is heavily imbalanced: paid games account for 79.4% of records and free games for only 20.6% (Figure 1). This means a naive majority-class classifier would already achieve ~79% accuracy, so model accuracy must be interpreted alongside sensitivity and specificity.

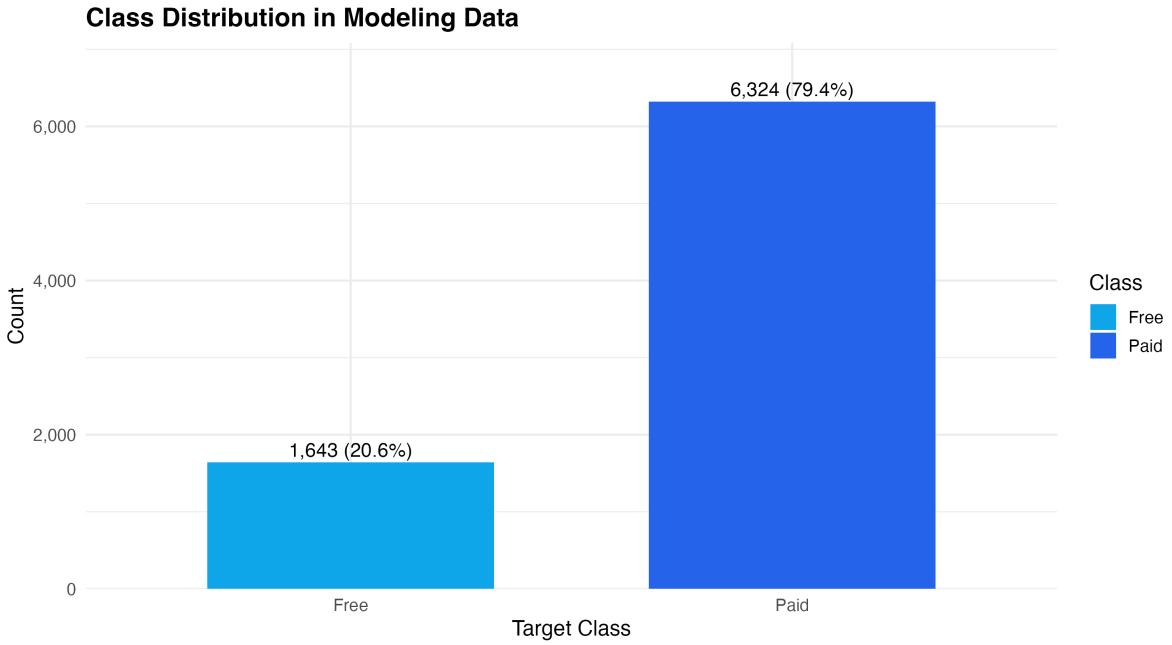

Figure 1: Class distribution of the target variable. Paid games outnumber free games roughly 4:1.

Figure 2 shows distributions of the four key numeric predictors. `n_categories` is right-skewed: most games belong to just a few categories. `platform_count` clusters at 1, reflecting that most titles are Windows-only. `release_year` rises sharply post-2015 as Steam's catalogue expanded. `required_age` is concentrated at 0, as most games carry no age restriction.

Figure 3 presents further feature relationships by class. Release-year distributions are similarly shaped for both classes, but paid games dominate throughout. The binary feature panel shows that paid games are far more likely to have DLC or a demo, consistent with using demos to drive paid conversions.

Figure 4 shows the difference in category prevalence between free and paid games. Single-player games strongly skew paid, while multiplayer and Steam leaderboard games tilt slightly toward free — consistent with free-to-play titles relying on large player pools for engagement.

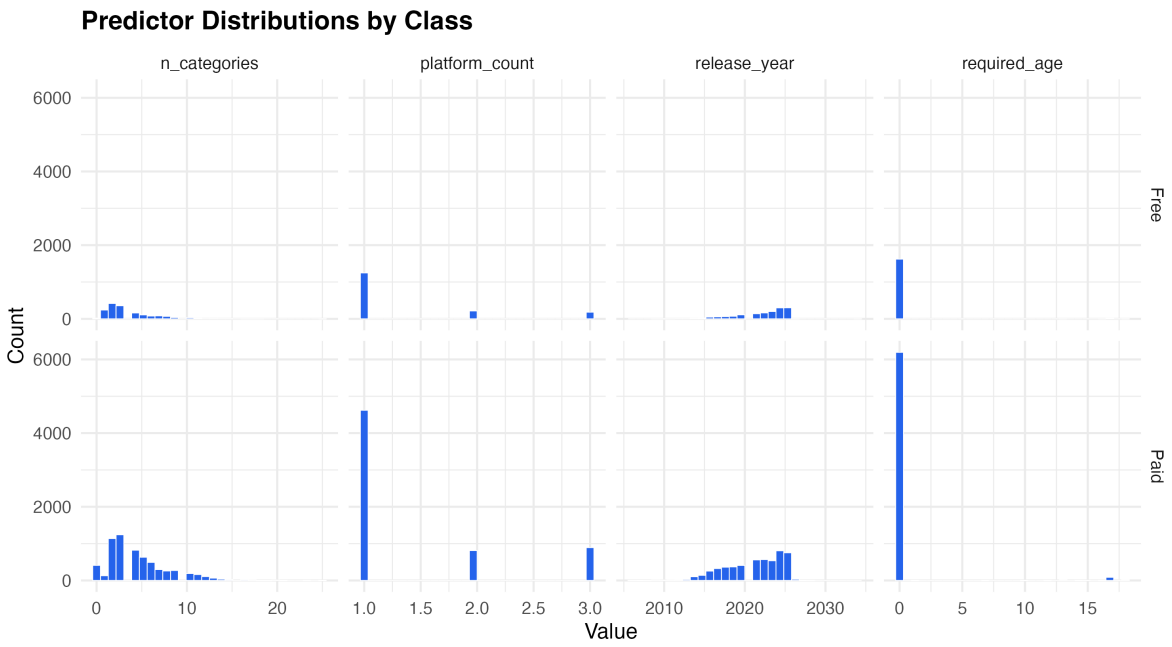

Figure 2: Distributions of the four key numeric predictors, faceted by free vs paid class.

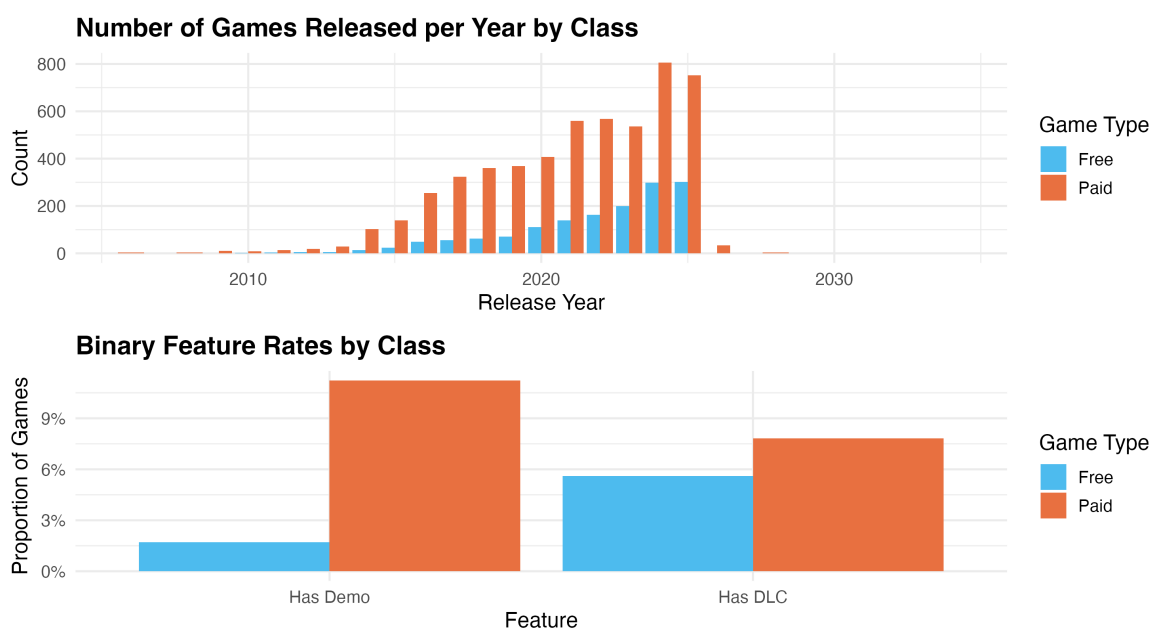

Figure 3: Release year trends and binary feature rates broken down by free vs paid status.

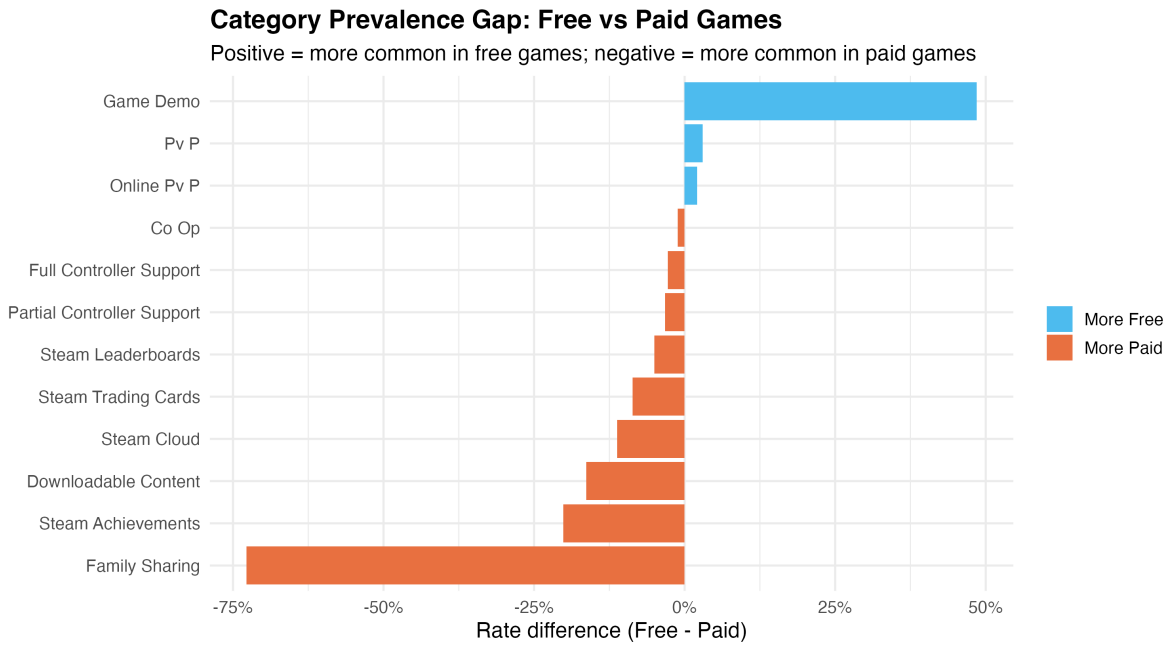

Figure 4: Difference in category prevalence between free and paid games.

Modelling

The dataset was split 80/20 into training and test sets using stratified sampling to preserve the class ratio. A majority-class baseline — predicting "Paid" for every observation — yields 79.4% accuracy.

Logistic regression was trained using `caret` (Kuhn (2024)) with 5-fold cross-validation optimising for ROC AUC. Given the imbalanced nature of the data, we chose ROC AUC as our evaluation metric, because it's suitable for evaluating model performance across all thresholds, and is less sensitive to class imbalance.

Only pre-launch observable features were included as predictors. We made this choice for two reasons. For one, the decision as to whether a game will be free or paid is generally made before launch; using only pre-launch features better reflects the reality that we are trying to model, and is more useful for making predictions before or during launch. For another, knowing that post-launch features have a chance of being correlated with cost (e.g. reviews may be more positive for free games; playtime may be higher for paid games), we wanted to avoid this "leakage".

Evaluation and Results

Classification metrics for the model on the held-out test set are shown in Table 1.

Table 1: Classification metrics for the logistic regression model on the held-out test set.

metric	value
Baseline accuracy (majority class)	0.7940
Model accuracy	0.9585
Cohen's Kappa	0.8667
Sensitivity (Free recall)	0.8323
Specificity (Paid recall)	0.9913
Balanced accuracy	0.9118
ROC AUC	0.9230

The model improves on the baseline by 16 percentage points. Specificity is near-perfect at 0.991, while sensitivity reaches 0.832. Cohen's Kappa of 0.867 confirms performance well beyond chance (Cadili, Fyson, and Widmann (2015)). Balanced accuracy of 0.912 indicates the model handles both classes reasonably despite the imbalance (Silva (2024)).

In Figure 5 we show the ROC curve, which hugs the top-left corner (AUC = 0.923), confirming strong discriminative ability across all decision thresholds.

The confusion matrix in Figure 6 shows the raw prediction counts. The model correctly identifies the vast majority of paid games, while missing a smaller proportion of free games due to class imbalance.

The top 15 variable importance scores are presented in Table 2. The five most important predictors are family sharing, single-player, Steam leaderboards, multiplayer, and demo availability.

Table 2: Top 15 variable importance scores from the logistic regression model.

term	Estimate	odds_ratio	Pr(>|z|)
game_typeseries	-39.3360210	0.0000000	0.9951821
game_typedemo	-32.9848849	0.0000000	0.9943019
game_typemod	-31.9277191	0.0000000	0.9953197
windows_supportTRUE	-16.6667177	0.0000001	0.9968516
game_typegame	-14.8650142	0.0000004	0.9974284
game_typeepisode	-14.6994101	0.0000004	0.9974571
game_typevideo	-13.5617122	0.0000013	0.9976539
game_typedlc	-13.0116513	0.0000022	0.9977491
game_typemusic	-12.5921752	0.0000034	0.9978216

Table 2: Top 15 variable importance scores from the logistic regression model.

term	Estimate	odds_ratio	Pr(>|z|)
cat_family_sharingTRUE	4.6434380	103.9009423	0.0000000
cat_single_playerTRUE	-2.2241078	0.1081639	0.0000000
cat_steam_leaderboardsTRUE	1.0063257	2.7355314	0.0001426
cat_multi_playerTRUE	-0.8675205	0.4199916	0.0007529
has_demoTRUE	0.7684477	2.1564162	0.0049835
linux_supportTRUE	-0.4929977	0.6107927	0.0176968

Discussion

The logistic regression classifier achieves strong overall performance (accuracy 0.959, AUC 0.923), but the asymmetry between sensitivity (0.832) and specificity (0.991) warrants careful interpretation. The model is more reliable at identifying paid games than free ones — a consequence of the 4:1 class imbalance in the training data.

The variable importance results are counterintuitive in places. Family sharing emerges as the single strongest predictor of a game being *free*, yet the EDA showed that paid games have family sharing enabled more often in absolute frequency. The logistic regression coefficient captures the conditional relationship after controlling for all other variables, which can reverse marginal associations — a known consequence of multivariate adjustment.

The strong influence of the single-player category on predicting *paid* games aligns with industry convention: narrative or campaign-driven titles typically require a purchase, whereas free-to-play games rely on multiplayer engagement and microtransactions.

An interesting follow-up analysis would be to define multiplayer sub-categories (MOBA, MMORPG, co-op) to determine whether finer genre distinctions improve predictive power, or to extend the target variable beyond binary free/paid to capture the full pricing distribution.

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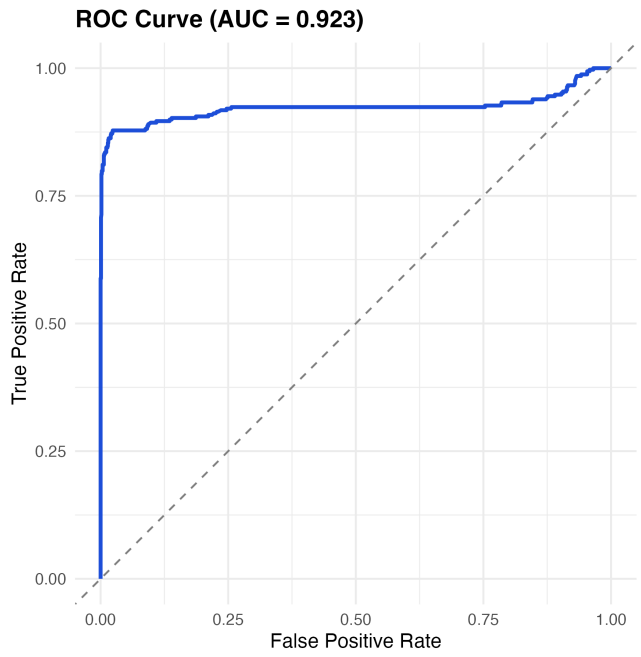

Figure 5: ROC curve on the held-out test set (AUC = 0.923).

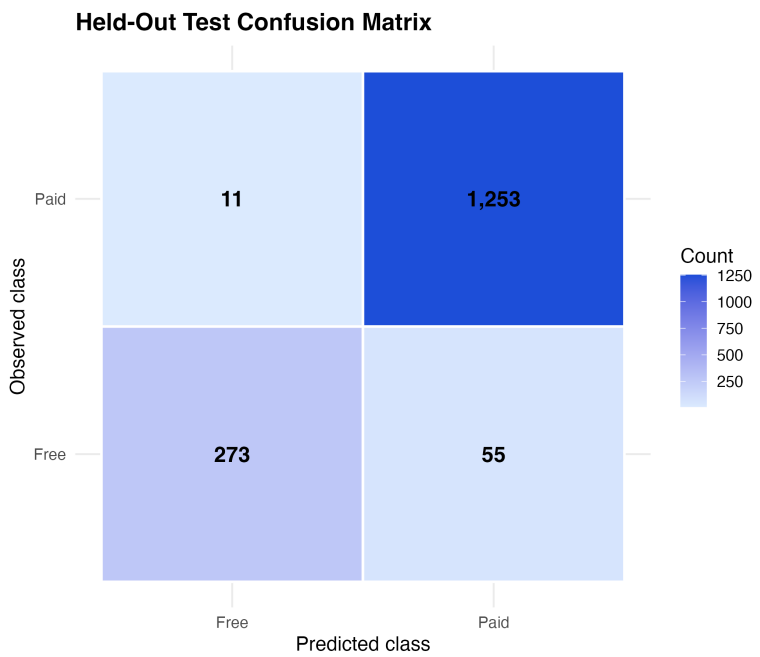

Figure 6: Confusion matrix on the held-out test set.

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